Figure 4. Enlargement extends polarity heading memory under fluctuating input

Model panels use positive lognormal input with mean f times the native reference: Wave-pinning 0.1, Holmes 1, Spring 4 and Goryachev 6.4. The Gaussian driving field has unit variance, lognormal amplitude σ = 1, spatial parameter 1.5 and OU correlation time 50 (Spring basal-physical 0.15 and 5). Spring input is stationary in material coordinate, so its physical correlation length scales with circumference. Native clocks are not calibrated between models. Unless stated otherwise, error bars are 95% whole-history bootstrap intervals.

A, Detector-free null kymographs at C = 20 and 60: identical physical diffusion, size-independent jumping, or perimeter-proportional jumping. Each facet shows the same activity measure.

B, Null heading autocorrelation; persistence size exponents are 2, 0 and −1. C = 20 is light and C = 60 dark.

C, Native-model kymographs at f = 0.03 (Wave-pinning), 0.02 (Holmes), 0.005 (Spring) and 0.01 (Goryachev). Arc position is centred at the initial heading; time is elapsed after burn-in. Goryachev activity ranges are labelled separately for each size.

D, Model heading autocorrelation at the inputs in C, for the smallest and largest rings (Spring C = 1 and 3). Solid curves show the smallest ring and dotted curves the largest. Endpoint eligibility requires polarity coherence $|P| \geq 0.1$. These direct curves show longer heading memory in larger rings.

E, Local persistence exponent α versus input fraction f, using three neighbouring sizes (two at endpoints), with whole-history 95% intervals. In E and G–I, sizes are C = 20, 30, 40, 50 and 60 (Spring 1, 1.5, 2, 2.5 and 3), light to dark. Open circles in D, E, G and I mark the declared combined forcing-control flags; unavailable rates are omitted.

F, Noise-tolerance fold at heading-correlation lag 2000 (Spring 200), relative to the smallest ring. Spring circumference is multiplied by 20 for display. The dashed grey guide is proportional to circumference.

G, Physical within-tenure drift coefficient normalized by the smallest size at the same input, using five-frame persistent dominant tenures. Goryachev uses a different ordinate range, 10^{-7}–2; the other models use 0.2–8.

H, Drift plus counted five-frame-dominant takeovers, divided by observed heading decorrelation rate, at every designed input. Solid lower and dashed upper limits bound every unscored takeover loss between 0 and 2. Shading denotes identification intervals, not confidence intervals. Grey line: unity. Open upward triangles with arrows mark an unbounded upper limit where drift is unavailable. Only the unknown drift contribution has lower bound zero; counted-takeover lower contributions are retained. Crosses at the top mark unavailable observed rates.

I, Signed remainder share versus input, using point regions with native-amplitude scaling. R includes amplitude, within-region shape, unresolved activity and background contributions; its sign and magnitude depend on the accounting convention. This convention retains measured native coherence, avoids the unit-modulus single-region normalization artefact and gives a common comparison across inputs and sizes. The decomposition is descriptive. Exact own-mask

Figure 4. Enlargement extends polarity heading memory under fluctuating input · continued

and normalization sensitivities appear in S4.5; complete-mask reconstruction is an algebraic identity, not causal validation.

J, Probability that a certified persistent secondary birth becomes globally dominant within 20 input correlation times, at each model's labelled input. Filled circles: 0 < d ≤ C/6; open squares: d > C/6, where d is estimated edge-to-edge birth–incumbent separation. Probabilities concern the certified positive-distance subset; a [0,1] interval is uninformative. Lower finite-horizon incidence can reflect slower or rarer takeover; lifetime takeover probability is unidentified. Input selection, certification denominators and interval procedures are in Methods.

Scope/limitations: The strong-input regime remains unexplained.

$$Q_t = \frac{\sum_{i \in I_t} A_{i,t} e^{i\theta_{i,t}}}{\sum_{i \in I_t} A_{i,t}}, \qquad r_t = |P_t|, \quad \widetilde{Q}_t = r_t Q_t \tag{4.1}$$

Q is the normalized point-region moment; P=M₁/M₀ is measured from the native field. Its amplitude r prevents the unit-modulus single-region normalization artefact.

$$\Delta P_j = \bar{r}\,\Delta Q_j \ (j = \mathrm{M}, \mathrm{A}, \mathrm{B}), \qquad \Delta P_{\mathrm{R}} = \Delta P - \bar{r}\,\Delta Q \tag{4.2}$$

The three ΔQ components use the symmetric Shapley convention for motion, coexisting amount and membership; r̄ is mean endpoint native amplitude. R closes the native moment change and includes amplitude variation, shape and background. This is an accounting convention, not causal attribution.

$$L_j = \frac{\mathrm{Re}\left[\Delta P_j \overline{(H_1 - H_0)}\right]}{|P_0| + |P_1|}, \quad H_t = \frac{P_t}{|P_t|}, \quad \sum_j L_j = 1 - \cos(\theta_1 - \theta_0) \tag{4.3}$$

Pooled signed fractions are sums of L_j divided by total measured heading loss. They retain cancellation, can exceed one, and depend on the stated accounting convention; they are not causal shares.

$$k(\ell) = -\frac{\ln C(\ell)}{\ell}, \qquad k \propto \bar{C}^{-\alpha} \tag{4.4}$$

C(ℓ) is activity-heading autocorrelation; mean C is actual circumference. Endpoint eligibility uses |P|≥0.1. Main H evaluates every designed input; unavailable observed rates remain explicit.

$$a_n(t) = \frac{1}{N} \sum_{j=1}^{N} u_j(t) e^{in\theta_j}, \qquad P(t) = \frac{a_1(t)}{a_0(t)}, \qquad q_n = \frac{2\pi n}{\bar{C}} \tag{4.5}$$

Figure 4. Enlargement extends polarity heading memory under fluctuating input · continued

Fourier modes use cell-centre material phases and the positive convention matching the native first moment. Heading is the phase of P; Spring physical wavenumber uses mean circumference.

$$\gamma_n = -\frac{\ln C_n(\ell_*)}{\ell_*}, \qquad k(C) = \gamma(2\pi/C), \qquad \alpha = \frac{\mathrm{d}\ln\gamma}{\mathrm{d}\ln q} \tag{4.6}$$

The last two equalities are the detector-free hypothesis under a common size-independent mode curve, not an identity assumed in analysis. Complex correlations use the nearest-half lag rule. Half-crossing status is exported; primary collapse and slope comparisons use observed crossings. The declared descriptive screens and stencil sensitivities are reported model by model.

$$k^{\mathrm{lo}}_{\mathrm{pred}} = q_1^2 D_s + \frac{\sum_{\mathrm{scored}} (1-\cos\Delta\theta)}{T}, \qquad k^{\mathrm{hi}}_{\mathrm{pred}} = k^{\mathrm{lo}}_{\mathrm{pred}} + \frac{2N_{\mathrm{unscored}}}{T} \tag{4.7}$$

All counted takeovers enter the interval: each unscored angular loss lies between 0 and 2. T is eligible exposure. Missing drift has a nonnegative lower bound and unbounded upper limit; an unavailable observed heading rate prevents a numerical ratio.